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Instruction-Grounded Visual Projectors for Continual Learning of Generative Vision-Language Models

Hyundong Jin[1] Hyung Jin Chang[2] Eunwoo Kim[1,*]
[1]School of Computer Science and Engineering, Chung-Ang University
[2]School of Computer Science, University of Birmingham
{jude0316, eunwoo}@cau.ac.kr h.j.chang@bham.ac.uk

Abstract

Continual learning enables pre-trained generative vision-language models (VLMs) to incorporate knowledge from new tasks without retraining data from previous ones. Recent methods update a visual projector to translate visual information for new tasks, connecting pre-trained vision encoders with large language models. However, such adjustments may cause the models to prioritize visual inputs over language instructions, particularly learning tasks with repetitive types of textual instructions. To address the neglect of language instructions, we propose a novel framework that grounds the translation of visual information on instructions for language models. We introduce a mixture of visual projectors, each serving as a specialized visual-to-language translation expert based on the given instruction context to adapt to new tasks. To avoid using experts for irrelevant instruction contexts, we propose an expert recommendation strategy that reuses experts for tasks similar to those previously learned. Additionally, we introduce expert pruning to alleviate interference from the use of experts that cumulatively activated in previous tasks. Extensive experiments on diverse vision-language tasks demonstrate that our method outperforms existing continual learning approaches by generating instruction-following responses.

1. Introduction

Generative Vision-Language Models (VLMs) [4, 33, 35, 55] have demonstrated remarkable performance in a variety of tasks, such as caption generation [24], visual question answering [42], and creative applications [1, 55]. Their impressive performance is attributable to the robust capabilities of large language models in conjunction with a translation module that enables these models to comprehend visual information from a visual encoder. Despite their remarkable performance, generative VLMs face significant challenges in adapting to newly emerging tasks. Retraining these models with pre-trained and newly acquired data imposes considerable computational costs due to the large volume of pre-trained data [26] and its limited public accessibility [40]. Furthermore, training models solely on new tasks leads to overwriting previously acquired knowledge, resulting in catastrophic forgetting [25, 37].

Continual learning methods [22, 23, 25, 31, 41, 48] aim to address these challenges by allowing models to incorporate new tasks while preserving old knowledge with access only to the data of newly emerging tasks. Existing continual learning methods rely on storing a small set of previous data and training it jointly with the new data [13, 41] or on regularizing parameter updates that are important for previous tasks [25]. Other methods [31, 44, 48] incorporate a small set of learnable parameters into pre-trained knowl-

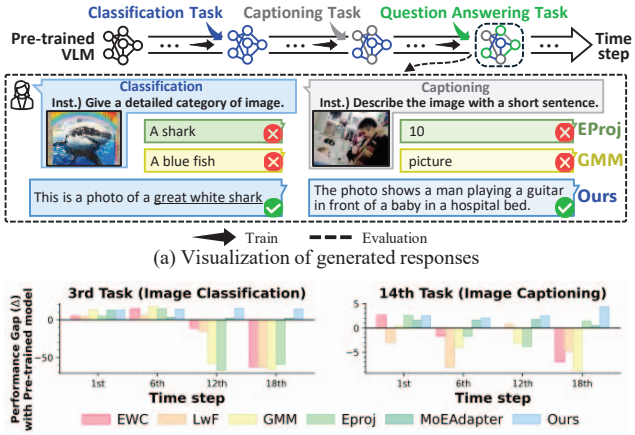


Figure 1. (a) Generated responses for classification and captioning tasks from continual learning methods for VLM after training on a question answering task. Existing methods, EProj [16] and GMM [4], neglect text instruction. (b) Performance gap between the pre-trained model and models trained on sequential tasks using various continual learning methods. The proposed method outperforms others in preventing knowledge forgetting.

*Corresponding author.

edge to capture task-specific knowledge. Despite these advancements, continual learning for VLMs remains underexplored. Recently, a few studies [4, 52] proposed to directly update the pre-trained visual projector of VLM that converts visual features for language models to learn new tasks, where tasks share similar instruction templates accompanied by similar answer types. Although these methods show promising results for such tasks, they neglect text instructions for response generation after training on tasks with varying instructions, as shown in Figure 1 (a).

In this paper, we propose a novel framework for transforming visual information grounded on language instructions for continual learning of VLMs. To this end, we introduce a **M**ixture of **V**isual **P**rojectors, named **MVP**, each serving as an expert dedicated to transforming visual information based on a given instruction context. In particular, our routing scheme dynamically adapts to both the semantic content of the instruction and the visual information, leveraging translation experts that align with the linguistic intent. To encourage the use of appropriate experts for given instructions, we propose an expert recommendation strategy that promotes reusing experts associated with relevant previous tasks while discouraging those specialized in other instructions. In addition, we propose expert pruning to mitigate negative transfer to future tasks induced by the reuse of redundantly activated experts in previous tasks. To preserve the zero-shot capabilities of pre-trained VLMs, we adaptively aggregate the transformed visual information from the proposed mixture of visual projectors and the pre-trained projector, calibrating their contributions based on the relevance of the given input data to the learned tasks.

To evaluate the proposed method, we perform vision-language tasks, including classification [17], captioning [39], and question answering [42], with a diverse range of visual and textual data. We evaluated its ability to retain new knowledge and maintain zero-shot performance, as illustrated in Figure 1 (b). The experimental results show that the proposed method outperforms its competitors by translating visual information for a given instructional context. The contributions of our work are four-fold:

- We propose a novel framework for adaptively translating visual information grounded in textual instruction for continual learning of generative vision-language models.
- We present expert recommendation and pruning strategies for sparse activation of experts from semantically relevant old tasks while mitigating negative knowledge transfer.
- To preserve zero-shot capabilities, we adaptively aggregate knowledge by balancing pre-trained and instruction context-aware projections.
- Extensive experiments show that the proposed framework outperforms other continual learning methods in preserving both pre-trained and newly learned knowledge.

2. Related Work

Generative Vision-Language Models (VLMs) [11, 29, 32, 49, 55] demonstrate notable zero-shot generalization and in-context learning capabilities for vision-language tasks. These models harness the robust capabilities of Large Language Models (LLMs) [9, 46] by employing connection modules that bridge the visual encoder with the LLM, e.g., Qformers [11, 29] or resampler [32, 49, 55]. By translating visual information into comprehensible representations for the language model, VLMs can interpret and generate responses based on complex visual and textual inputs. Updating the connection module can directly improve the visual information for the language model with negligible parameter updates compared to updating the visual encoder and the language model [5, 55]. However, this approach is vulnerable in that previously acquired information can be significantly compromised during adaptation to new vision-language tasks. Therefore, a central challenge in VLMs is preserving the ability to follow previously learned instructions while adapting to newly introduced ones.

Continual Learning aims to learn from a stream of tasks without forgetting previously acquired knowledge [25, 37]. Existing continual learning methods incorporate additional regularization terms to penalize changes in critical parameters to maintain performance on earlier tasks [3, 6, 25, 51], or replay a memory buffer containing a small set of data from previous tasks when learning new tasks [13, 34, 41]. Continual learning with parameter-efficient tuning methods [15, 44, 47, 48] learns task-specific representations by selectively updating a minimal subset of parameters [19, 21]. Recently, several approaches for continual learning in visual language models [4, 52] incorporate a visual projector that is shared across tasks. Although these approaches might be beneficial for tasks that share similar instruction templates, learning tasks that include contextually different instructions might cause the model to easily forget the previously learned visual representations. In contrast, the proposed approach addresses this limitation by leveraging context-aware visual-to-language translation for given textual instructions.

Mixture-of-Experts (MoE) [8, 20, 36, 38, 50] consists of multiple distinct networks, called experts, and a routing network to select appropriate experts and aggregate the outputs of the selected experts. In continual learning, MoE-based methods have been introduced [2, 8, 50] by expanding the router [2, 8, 50] or adding more experts [2, 8] in response to new tasks. Additionally, existing methods [8, 50] prohibit updates to a subset of experts associated with old tasks to mitigate forgetting when learning new tasks. However, this can potentially hinder learning for new tasks by freezing important experts, and a large number of frozen experts complicates capturing new representations. In this work, we resolve these limitations by recommending experts to

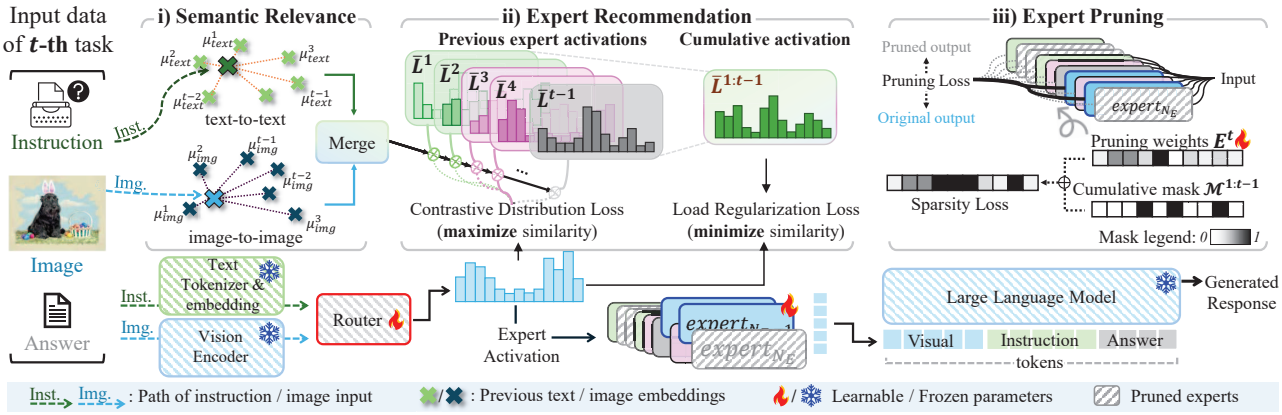


Figure 2. An illustration of the proposed continual learning framework for the t-th vision-language task. A router determines projector (expert) activations based on the embeddings of the instruction and image. To prevent detrimental expert activation of a router, **(i)** we first evaluate the semantic relevance between the current and previous tasks within an embedding space. **(ii)** We then encourage the activation of experts used in semantically related previous tasks for new tasks, while suppressing those with high cumulative activation frequencies. **(iii)** To mitigate negative knowledge transfer to subsequent tasks, we prune and reinitialize experts with parameters of the pre-trained projector, thus securing learning capacity for subsequent tasks. Finally, the large language model generates responses through vision, instruction, and answer tokens.

be updated based on their relevance to the new task while preventing reliance on irrelevant ones, ensuring that experts remain active to provide sufficient learning capacity for new tasks. In addition, we propose expert pruning to break the cycle of negative knowledge transfer caused by regularizing balanced expert utilization, which risks subsequent tasks reusing redundantly updated experts from previous tasks.

3. Methodology

3.1. Overview

Continual learning aims to enable pre-trained generative vision-language models (VLMs) to preserve previously acquired knowledge while sequentially adapting to a sequence of novel tasks. At time step t, VLM is trained on the t-th vision-lanugage task, where the t-th task $T^t = \{x_i^t, q_i^t, y_i^t\}_{i=1}^{N^t}$ comprises N^t triplets, where x_i^t, q_i^t, and y_i^t represent an image, a text instruction, and a text-based answer for the i-th sample, respectively. For each triplet, the image feature $\boldsymbol{x}_{i,\text{img}}^t = \mathcal{F}_v(x_i^t)$ is extracted by the vision encoder $\mathcal{F}_v$, and the instruction tokens $\boldsymbol{x}_{i,\text{text}}^t = \mathcal{F}_e(q_i^t)$ and the answer tokens $\boldsymbol{y}_i^t = \mathcal{F}_e(y_i^t)$ are extracted by the text embedding function $\mathcal{F}_e$ of the pre-trained model.

To make the image feature $\boldsymbol{x}_{i,\text{img}}^t$ compatible with the language model, a visual projector $\mathcal{V}$ transforms it into $\bar{\boldsymbol{x}}_{i,\text{img}}^t = \mathcal{V}(\boldsymbol{x}_{i,\text{img}}^t)$. Then the auto-regressive decoder $\mathcal{D}$ generates responses by receiving the tokens $\bar{\boldsymbol{x}}_{i,\text{img}}^t$, $\boldsymbol{x}_{i,\text{text}}^t$, and $\boldsymbol{y}_i^t$. A naïve method to learn a new task is to update the pre-trained visual projector to align the predicted responses with the answer tokens by minimizing the cross-entropy loss as $\mathcal{L}_{ce} = -\sum_{i=1}^{N^t}\sum_{j=1}^{L} \boldsymbol{y}_{i,j}^t \log \hat{\boldsymbol{y}}_{i,j}^t$, where L denotes the number of tokens in the response. $\boldsymbol{y}_{i,j}^t$ denotes the j-th token of the answer for the i-th sample, and $\hat{\boldsymbol{y}}_{i,j}^t$ denotes the corresponding predicted token.

Although updating the visual projector of a VLM is a parameter-efficient approach [4, 16], *training it with similar repetitive instructions restricts its translation ability to reflecting only those instruction contexts. This results in the model neglecting different text instructions*, as illustrated in Figure 1 (a). To address this, we propose a novel framework that adaptively translates visual information via a mixture of projectors to provide instruction-context aware visual translation. To encourage the use of projectors for relevant contexts, we introduce a recommendation strategy that promotes the activation of projectors associated with semantically similar previous tasks. Additionally, we present expert pruning to eliminate redundant experts that may cause negative transfer to subsequent tasks. The overall framework is illustrated in Figure 2.

3.2. Mixture-of-Visual Projectors

We present a method to adaptively adjust visual projectors grounded on the text instructions of a new task. Our framework includes a router $\mathcal{R}$ and a set of projectors (experts) $\{\mathcal{E}_j\}_{j=1}^{N_E}$, each contributing as a visual-to-language translator for the given instructional context. For the t-th task, the router takes in multimodal data $\boldsymbol{x}_{i,\text{img}}^t$ and $\boldsymbol{x}_{i,\text{text}}^t$ to select the appropriate experts for translating $\boldsymbol{x}_{i,\text{img}}^t$ for a language model. The aggregated output from the selected experts is

$$\tilde{\boldsymbol{x}}_{i,\text{img}}^t = \sum_{j=1}^{N_E} w_j \mathcal{E}_j(\boldsymbol{x}_{i,\text{img}}^t), \tag{1}$$

where w_j denotes the routing weight determined by $\mathcal{R}$, which quantifies the contribution of the j-th expert. To ensure the visual projector grounded on the text instructions of newer tasks, we define the weights $W = \{w_j\}_{j=1}^{N_E}$ to be conditioned on the instruction tokens as

$$W = \text{Softmax}(\text{Top-}K(\mathcal{R}(\boldsymbol{x}^t_{i,\text{img}}, \boldsymbol{x}^t_{i,\text{text}}))), \tag{2}$$

where K denotes the pre-defined number of experts engaged for the translation. To provide the language model with a richer representation, we integrate the translated token from the newly selected experts with the token obtained from the pre-trained projector $\mathcal{V}$ as

$$\tilde{\boldsymbol{x}}^t_{i,\text{img}} = \frac{1}{K+1}\left(\mathcal{V}(\boldsymbol{x}^t_{i,\text{img}}) + \sum_{j=1}^{N_E} w_j \mathcal{E}_j(\boldsymbol{x}^t_{i,\text{img}})\right). \tag{3}$$

3.2.1. Expert Recommendation

To leverage the experts associated with old tasks that share similar visual-linguistic contexts with the new task, we introduce a recommendation strategy driven by the semantic relevance between old and new tasks. Specifically, we compute similarity scores for vision and language instructions: $\boldsymbol{S}^t_{\text{img}} = \{s^{t'}_{\text{img}}\}_{t'=1}^{t-1}$, which measures the cosine similarity between $\boldsymbol{x}^t_{i,\text{img}}$ and the stored average image embedding $\mu^{t'}_{\text{img}}$ of the t'-th previous task. Similarly, the corresponding set of textual similarity scores, $\boldsymbol{S}^t_{\text{text}} = \{s^{t'}_{\text{text}}\}_{t'=1}^{t-1}$, is computed from the current input $x^t_{i,\text{text}}$ and the stored instruction embedding $\mu^{t'}_{\text{text}}$ [1]. These are combined as $\boldsymbol{S}^t = \{s^{t'}\}_{t'=1}^{t-1}$, where

$$s^{t'} = \alpha \cdot \sigma(s^{t'}_{\text{img}}) + (1-\alpha)\cdot\sigma(s^{t'}_{\text{text}}). \tag{4}$$

$\sigma(\cdot)$ is the sigmoid function, and α balances the two scores. A higher score $s^{t'}$ indicates stronger alignment with the t'-th previous task.

Given these scores, we leverage a contrastive distribution loss [45] to encourage the outputs of the router to associate activation patterns of similar tasks while differentiating them from dissimilar ones. Let us denote the record of expert activation frequency from previous tasks by $\bar{L}^{t'}$ and the router output for the current task by L^t, both normalized via a Softmax function. The recommendation loss becomes $\mathcal{L}_{rec} = \sum_{i=1}^{N^t} H(\boldsymbol{S}^t, \pi_{t,t'})$, where

$$\pi_{t,t'} = \frac{\exp(\text{sim}(\bar{L}^{t'}, L^t))}{\sum_{t'=1}^{t-1}\exp(\text{sim}(\bar{L}^{t'}, L^t)/\tau)}, \tag{5}$$

$H(\cdot)$ is the cross-entropy, $\text{sim}(\cdot)$ is the cosine similarity, and τ is the temperature parameter. By emphasizing the expert patterns that correlate with previously encountered tasks, this recommendation encourages the router to reuse relevant expertise while minimizing reliance on less helpful ones.

Activation Bias Reduction. Although a mixture of experts provides flexibility in adapting to different tasks, they can lead the router to assign large weights to only a few experts [27, 43]. To mitigate such concentration, we introduce an additional regularization term that promotes diverse expert engagement by suppressing the activation of those specialized in previous tasks. We minimize

$$\mathcal{L}_{bias} = \frac{1}{2}\left(1 + \frac{\langle \bar{L}^{1:t-1}, L^t\rangle}{\|\bar{L}^{1:t-1}\|_2 \cdot \|L^t\|_2}\right), \tag{6}$$

where $\bar{L}^{1:t-1}$ is the normalized aggregated historical activation vector across all previous tasks. $\langle\cdot,\cdot\rangle$ and $\|\cdot\|_2$ denote the inner product and the ℓ_2 norm, respectively. This promotes the activation of a diverse set of unused experts while aligning the activations of new tasks with those from previous related tasks. To summarize, the proposed framework updates $\mathcal{R}$ and $\{\mathcal{E}_j\}_{j=1}^{N_E}$ by minimizing $\mathcal{L} = \mathcal{L}_{ce} + \lambda_{rec}\mathcal{L}_{rec} + \lambda_{bias}\mathcal{L}_{bias}$, where λ_{rec} and λ_{bias} are weighting factors for the respective losses.

3.2.2. Expert Pruning

A large number of previously activated experts may cause negative knowledge transfer to subsequent tasks. To alleviate the interference, we propose expert pruning on a new task after training with $\mathcal{L}$. We introduce a learnable vector, $E^t = \{e^t_1, ..., e^t_{N_E}\}$, which is updated to limit the number of experts used for both the new and previous tasks. To discard redundant experts of the current task with sparsely activated experts, we minimize

$$\min_{E^t}\left\|\sum_{j=1}^{N_E}(w_j - e^t_j)\mathcal{E}_j(x^t_{i,\text{img}})\right\|_F + \|\mathcal{M}^{1:t-1} + E^t\|_1, \tag{7}$$

where $\|\cdot\|_F$ and $\|\cdot\|_1$ denotes the Frobenius and the ℓ_1 norm. By minimizing the discrepancy between the output of the original expert mixture and its pruned counterpart, while constraining activation to a few experts, we identify a sparse subset of experts for the current task. The updated E^t is thresholded to produce a binary mask $\mathcal{M}^t$. After pruning the experts by the accumulated binary mask $\mathcal{M}^{1:t}$, the router is then fine-tuned to leverage the survived projectors for previous and new tasks. To collect inputs for router fine-tuning, we sample from a normal distribution defined by the stored mean and covariance of images and text instructions from observed tasks $t' \in [1, t]$. For each collected data, the corresponding label is sampled from $\text{Softmax}(L^{t'} \odot \mathcal{M}^{t'})$ over the indices of experts $1, 2, \ldots, N_E$. We minimize the cross-entropy loss to fine-tune the router.

[1] $\mu^{t'}_{\text{img}} = \text{Average}(\{x^{t'}_{i,\text{img}}\}_{i=1}^{N^{t'}})$ and $\mu^{t'}_{\text{text}} = \text{Average}(\{x^{t'}_{i,\text{text}}\}_{i=1}^{N^{t'}})$

3.3. Adaptive Knowledge Aggregation

Preserving the generalization capability of pre-trained models developed on extensive datasets remains critical. Our framework can effectively maintain this capability, as the proposed mixture of visual projectors is residually connected to the pre-trained projector. To this end, we present an adaptive knowledge aggregation strategy in the inference phase by leveraging the contribution of the MoE branch based on the semantic relevance of the data to previously encountered information.

Specifically, after learning the t-th task, we have the stored average visual and instruction embeddings $\{\mu_{\text{img}}^{t'}, \mu_{\text{text}}^{t'}\}_{t'=1}^{t}$. The semantic relevance between the inference data and the stored embeddings is calculated as Eq.(4), yielding a combined set of scores $\boldsymbol{S}_{\text{inf}}^{t} = \{s_{\text{inf}}^{t'}\}_{t'=1}^{t}$. Intuitively, a high relevance score $s_{\text{inf}}^{t'}$ indicates that the given vision-instruction paired data is more likely associated with the t'-th task. Consequently, the translated visual token for the language model is represented as a linear combination of the pre-trained projection and the MoE-translated projection, weighted by $\lambda_{\text{inf}} = \text{argmax}_{t' \in [1,t]} \{s_{\text{inf}}^{t'}\}_{t'=1}^{t}$, as

$$\frac{1}{\lambda_{\text{inf}} \cdot K + 1} \left(\mathcal{V}(\boldsymbol{x}_{i,\text{img}}^{\text{inf}}) + \lambda_{\text{inf}} \sum_{j=1}^{N_E} w_j \mathcal{E}_j(\boldsymbol{x}_{i,\text{img}}^{\text{inf}}) \right), \quad (8)$$

where $\boldsymbol{x}_{i,\text{img}}^{\text{inf}}$ denotes an input for inference and the weighting factor λ_{inf} ranges in $[0, 1]$. This adaptive aggregation ensures that the model retains its inherent zero-shot generalization capacity for unseen samples by calibrating the contributions of MoE-derived knowledge.

4. Experiments

4.1. Experimental Setting

Datasets. We applied the proposed **M**ixture-of-**V**isual **P**rojectors, named **MVP**, to continual learning of diverse vision-language tasks; image classification [17], image captioning [39], and image question answering [42]. For classification, we employed the ImageNet-R dataset [17] widely used in continual learning [44, 47]. The dataset was divided into 10 subsets containing 20 classes each, following the previous practices [44, 47]. We used Flickr-30K [39] for captioning tasks. We divided the dataset into four subsets based on the type of visual categories [12]: animals, instruments, scenes, and vehicles. For question answering, we employed the COCO-QA dataset [42]. We divided the dataset into four distinct subsets based on the type of language instructions [53]: object, count, color, and position. We define each subset as a task and train one task at each time step, which results in a sequence of 18 tasks.

Evaluation metrics. To evaluate the proposed method, we employ three metrics [50, 54]: *Last*, *Transfer*, and *Avg*. *Last* calculates the final performance after training all tasks. *Transfer* measures the models' zero-shot performance on unseen tasks. *Avg* is the average performance across all time steps. For classification and question answering tasks, we reported the metrics based on the accuracy of generated responses. For captioning tasks, the performance is evaluated by the RefCLIP-Score [18]. To enhance interpretability, we scaled the score, which ranges from 0 to 1. Please refer to the supplementary materials for measuring the accuracy of the generated responses.

Implementation details. We adopted Vicuna-7B [9] as the language decoder and employed the pre-trained ViT-g/14 [14] in conjunction with the Qformer as the visual encoder, following the practice in [4]. Additionally, we employed another VLM that utilized LLaMa-2-7B [46] as the language decoder and the visual encoder [14] without Qformer. We initialized the experts of the proposed method using the pre-trained visual projectors of VLMs, provided in [55]. We set the number of experts N_E to 20, K to 2, semantic score parameter α to 0.3, and the loss weights λ_{rec} and λ_{bias} to 1 for all experiments. The proposed method was trained using the Adam optimizer with β_1 and β_2 of 0.9 and 0.999. All experiments were conducted using NVIDIA RTX 3090 GPUs. For additional details, please refer to the supplementary material.

Comparison methods. To evaluate the proposed method, we compared it with existing continual learning methods, including conventional methods, LwF [30], EWC [25], and MoEAdapter [50], and the methods for VLMs, GMM [4] and EProj [16]. All comparison methods employed the same pre-trained VLMs and froze the visual and language encoders while training the visual projector for tasks. For the MoE-based continual learning method [50], we utilized its routing scheme and set its expert as the visual projector. Additionally, we reported the zero-shot performance of the pre-trained model, referred to as Zero-Shot.

4.2. Comparison Results

We evaluated the proposed method and the comparison methods on diverse vision-language tasks learned in the order of classification, captioning, and question answering. Table 1 shows the experimental results of continual learning methods using Vicuna as a large language model. Overall, the proposed method outperforms other continual learning methods in terms of the *Last*, *Avg*, and *Transfer* metrics across most tasks. The existing methods, LwF, EWC, GMM, and EProj, show unsatisfactory performance measured with *Last* due to overwriting previously learned knowledge in a shared projector across tasks by the latest learned question answering task. The proposed method, MVP, achieves superior classification performance, with notable performance gaps ranging from 12.80% to 75.71% compared to EProj and MoEAdapter. These results high-

Table 1. Results of the compared methods that employed Vicuna were measured using the *Last*, *Avg*, and *Transfer* metrics. The classification results were averaged across tasks. The names of the captioning and question answering tasks are abbreviated based on their visual and instruction types, respectively.

Method	Classification (Accuracy)			Captioning (RefCLIP-Score)				Question Answering (Accuracy)			
	T^1-T^3	T^4-T^6	T^7-T^{10}	Anim.	Inst.	Scen.	Vehi.	Obje.	Coun.	Colo.	Posi.
Vicuna											
Zero-Shot	67.79	63.16	62.87	77.08	75.24	74.24	73.43	68.52	42.55	62.62	32.79
Last											
LwF [30]	4.38	4.77	5.83	70.16	70.31	70.15	68.60	70.69	65.45	76.46	59.51
EWC [25]	5.61	6.80	5.83	67.00	67.39	66.13	66.47	65.72	63.63	73.69	44.13
GMM [4]	1.33	0.94	1.62	62.96	61.08	63.73	64.70	42.29	46.23	45.11	25.74
EProj [16]	8.50	12.82	11.81	73.78	71.43	72.39	72.23	39.77	60.00	65.84	24.69
MoEAdapter [50]	70.01	65.10	65.55	76.53	76.23	76.43	73.92	66.76	42.18	41.53	38.86
MVP (Ours)	**82.81**	**85.34**	**87.52**	**79.60**	**77.57**	**76.06**	**77.79**	**79.26**	**74.54**	**80.30**	**71.25**
Avg											
LwF [30]	48.37	49.42	51.62	70.63	70.34	68.56	69.18	53.99	47.21	60.22	37.51
EWC [25]	55.21	50.13	52.74	72.18	72.72	71.86	71.64	52.56	48.74	56.45	27.08
GMM [4]	43.92	42.08	50.70	69.62	69.15	68.84	68.80	48.39	48.18	48.47	27.71
EProj [16]	49.18	45.55	48.23	73.78	71.43	72.39	72.23	40.06	**50.88**	50.65	21.68
MoEAdapter [50]	71.50	67.16	66.27	76.65	75.97	74.40	74.66	67.95	42.60	53.82	36.48
MVP (Ours)	**81.34**	**82.03**	**85.07**	**78.71**	**76.92**	**75.92**	**76.25**	**70.16**	50.68	**71.34**	**38.55**
Transfer											
LwF [30]	69.68	64.30	71.26	67.15	67.97	66.74	68.41	49.07	43.80	58.47	36.17
EWC [25]	71.85	66.81	75.69	74.00	72.85	72.65	72.63	48.57	45.33	54.20	26.07
GMM [4]	74.59	65.82	81.96	70.40	70.30	69.84	69.77	42.29	46.23	45.11	25.74
EProj [16]	71.85	66.81	75.76	73.98	72.34	72.06	71.86	42.17	**48.99**	48.65	21.50
MoEAdapter [50]	75.01	66.61	67.05	76.80	75.80	74.38	74.94	68.29	42.69	55.35	36.34
MVP (Ours)	**77.95**	**73.34**	**82.52**	**78.02**	**76.88**	**75.71**	**75.91**	**67.52**	45.89	**70.17**	**36.62**

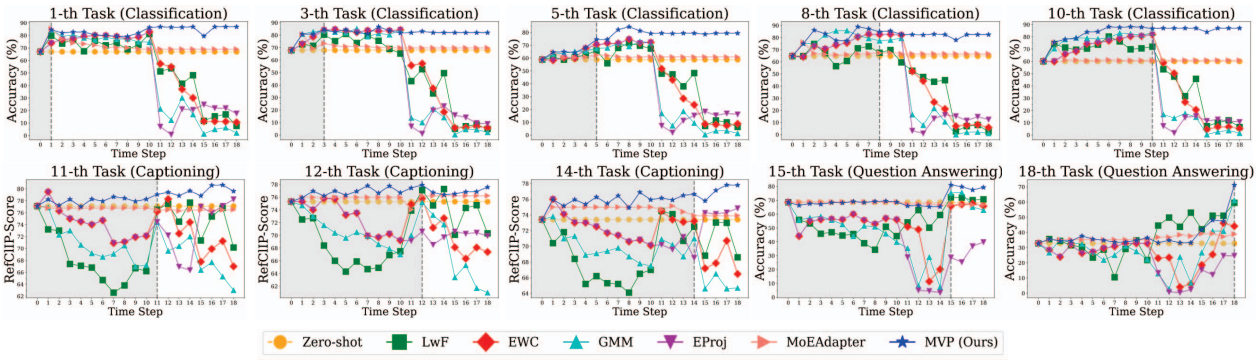


Figure 3. Performance of each task across time steps. The gray area represents the time steps preceding training for specific tasks, and the vertical dashed line indicates the time step at which the corresponding task is learned. Best viewed in color (×2).

light the importance of instruction context-aware translation in continual learning for VLMs.

Note that most compared methods using a shared visual projector, which tends to overfit on classification, achieve higher performance than Zero-Shot in *Transfer* but exhibit a significant performance drop in captioning and question answering tasks. In contrast, MVP achieves higher performance than Zero-Shot in most tasks, demonstrating that the relevance-based expert recommendation and knowledge aggregation contribute to effective context-aware translation. For the results of a different task order, please refer to the supplementary material.

We also present the performance of tasks at each time in Figure 3. As shown in the figure (top), most methods show improved performance on subsequent classification tasks after training on the other classification tasks. However, their performance significantly degrades after training on captioning and question answering tasks, with results lower than the zero-shot performance of the pre-trained model (indicated by the yellow horizontal line). The performance of captioning tasks decreases when newer tasks are introduced, with this decline becoming more pronounced following the introduction of the question answering tasks. In contrast, the proposed approach does not exhibit a performance drop in each task after learning other tasks. Notably, the performance of captioning tasks shows a slight improvement after learning classification tasks. This is attributed to the instruction-grounded translator that leverages knowledge of classification to help perform captioning tasks.

To evaluate the proposed method based on another VLM,

Table 2. Results of using VLM with LLaMa-2. Δ indicates the performance gap with the pre-trained VLM. Average indicates the average performance of tasks.

Method	Classification		Captioning		Question Answering	
	Average	Δ (↑)	Average	Δ (↑)	Average	Δ (↑)
LLaMa-2						
Zero-Shot	59.21	0.00	75.90	0.00	60.31	0.00
Last						
LwF [30]	70.13	+10.92	77.12	+1.22	54.79	-5.52
EWC [25]	52.55	-6.66	68.87	-7.03	31.60	-28.71
GMM [4]	21.28	-37.93	78.92	+3.02	59.48	-0.83
EProj [16]	53.68	-5.53	66.42	-9.48	30.44	-29.87
MoEAdapter [50]	61.30	+2.09	75.87	-0.03	59.26	-1.05
MVP (Ours)	**79.99**	**+20.78**	**78.72**	**+2.82**	**67.46**	**+7.15**
Avg						
LwF [30]	68.02	+8.81	73.50	-2.40	49.92	-10.39
EWC [25]	48.91	-10.30	65.57	-10.33	35.31	-25.00
GMM [4]	64.20	+4.99	73.31	-2.59	50.13	-10.18
EProj [16]	49.74	-9.47	64.61	-11.29	32.82	-27.49
MoEAdapter [50]	62.77	+3.56	75.87	-0.03	60.16	-0.15
MVP (Ours)	**78.58**	**+19.37**	**78.37**	**+2.47**	**61.66**	**+1.35**
Transfer						
LwF [30]	63.88	+4.67	71.81	-4.09	49.07	-11.24
EWC [25]	47.33	-11.88	64.24	-11.66	37.36	-22.95
GMM [4]	49.74	-9.47	64.61	-11.29	32.82	-27.49
EProj [16]	47.11	-12.10	63.92	-11.98	34.19	-26.12
MoEAdapter [50]	65.15	+5.94	75.08	-0.82	56.27	-4.04
MVP (Ours)	**74.39**	**+15.18**	**77.67**	**+1.77**	**57.16**	**-3.15**

we conducted additional experiments using LLaMa-2 [46], which has a different language decoder and a vision encoder compared to the previous VLM using Vicuna. Table 2 shows the results of the methods with the same task order as in Table 1. The trends of the results are similar to those in Table 1. Most competitors underperform Zero-Shot in *Last*, whereas MVP outperforms Zero-Shot by large average margins of 20.78%, 2.82, and 7.15% in classification, captioning, and question answering, respectively. MVP also outperforms the runner-up method, MoEAdapter, in *Transfer* by 9.24%, 2.59, and 0.89% across the tasks. This results from relevance-based calibration of newly acquired and pre-trained knowledge, unlike MoEAdapter, which relies solely on pre-trained knowledge for unseen tasks.

4.3. Ablation Study and Analysis

Ablation study. We conducted an ablation study by sequentially removing key components: visual projectors, recommendation loss $\mathcal{L}_{rec}$, activation bias reduction loss $\mathcal{L}_{bias}$, expert pruning, and adaptive knowledge aggregation (AKA). Table 3 shows the results of the ablation study. The method without AKA degrades not only *Transfer* but also *Last* compared to that with AKA, demonstrating the importance of calibrating learned knowledge based on semantic relevance. The method without pruning degrades the average performance of captioning (76.70 to 71.55) and question answering (76.17% to 75.54%) in *Last*, indicating that pruning helps to mitigate negative transfer by eliminating redundant experts. We also observe that the method without AKA and pruning shows slightly lower performance in

Table 3. Ablation study of MVP with different components.

Method	Classification		Captioning		Question Answering	
	Average	Δ (↑)	Average	Δ (↑)	Average	Δ (↑)
Vicuna						
Zero-shot	64.41	0.00	75.00	0.00	51.62	0.00
Last						
MVP (Complete method)	85.87	+21.46	77.75	+2.75	76.34	+24.72
w/o AKA	85.53	+21.12	76.70	+1.70	76.17	+24.55
w/o AKA, Prune	85.92	+21.51	71.55	-3.45	75.54	+23.92
w/o AKA, Prune, $\mathcal{L}_{bias}$	3.31	-61.10	70.70	-4.30	73.99	+22.37
w/o AKA, Prune, $\mathcal{L}_{bias}$, $\mathcal{L}_{rec}$	0.25	-64.16	68.51	-6.49	74.53	+22.91
w/o all (Single projector [4])	1.49	-62.92	63.10	-11.90	60.15	+8.53
Avg						
MVP (Complete method)	83.28	+18.87	76.94	+1.94	57.68	+6.06
w/o AKA	83.02	+18.61	72.21	-2.79	48.64	-2.98
w/o AKA, Prune	83.51	+19.10	69.22	-5.78	47.05	-4.57
w/o AKA, Prune, $\mathcal{L}_{bias}$	69.59	+5.18	71.60	-3.40	51.68	+0.06
w/o AKA, Prune, $\mathcal{L}_{bias}$, $\mathcal{L}_{rec}$	58.51	-5.90	70.59	-4.41	51.39	-0.23
w/o all (Single projector [4])	46.97	-17.44	69.10	-5.90	43.10	-8.52
Transfer						
MVP (Complete method)	78.45	+14.04	76.16	+1.16	50.89	-0.73
w/o AKA	78.40	+13.99	68.92	-6.08	44.55	-7.07
w/o AKA Prune	78.34	+13.93	68.78	-6.22	43.03	-8.59
w/o AKA, Prune, $\mathcal{L}_{bias}$	78.06	+13.65	71.47	-3.53	47.94	-3.68
w/o AKA, Prune, $\mathcal{L}_{bias}$, $\mathcal{L}_{rec}$	78.26	+13.85	70.45	-4.55	47.68	-3.94
w/o all (Single projector [4])	76.31	+11.90	69.97	-5.03	39.03	-12.59

Transfer than the approach that also removes $\mathcal{L}_{bias}$, as a diverse set of activated experts for seen tasks is repurposed for unseen tasks. However, the methods using $\mathcal{L}_{bias}$ yield substantial improvement compared to the method without it in *Last*, especially in classification (3.31% to 85.92%), by promoting diversified experts utilization. Additionally, the methods without $\mathcal{L}_{rec}$ shows performance drop measured in *Avg*, particularly in classification (Δ +5.18 to -5.90) and question answering (Δ +0.06 to -0.23).

Qualitative results. To investigate the changes in responses throughout the continual learning process, we provide a qualitative analysis. Figure 4 shows generated responses over time steps from different continual learning methods using Vicuna. The proposed method generates responses that reflect the given instruction for the visual input while maintaining the response quality similar to zero-shot performance, even before learning the task associated with the given image-instruction pair. In contrast, LwF and GMM often neglect the language instruction and forget previously learned detailed visual information, generating coarse-grained responses (e.g., "a drawing" for classification and "car" for captioning as shown in Figure 4). By leveraging the instruction-aware visual projectors, MVP generates specialized responses for learned tasks and responses similar to zero-shot performance for unseen tasks.

Expert activation frequency. We further analyze the combinations of three components, recommendation loss, activation bias reduction loss, and expert pruning, that influence the expert activation pattern, as in Figure 5. When only the recommendation loss is applied, a limited subset of experts is predominantly activated (similar observations in [27, 43]). In contrast, combining recommendation with the activation bias reduction loss promotes broader expert

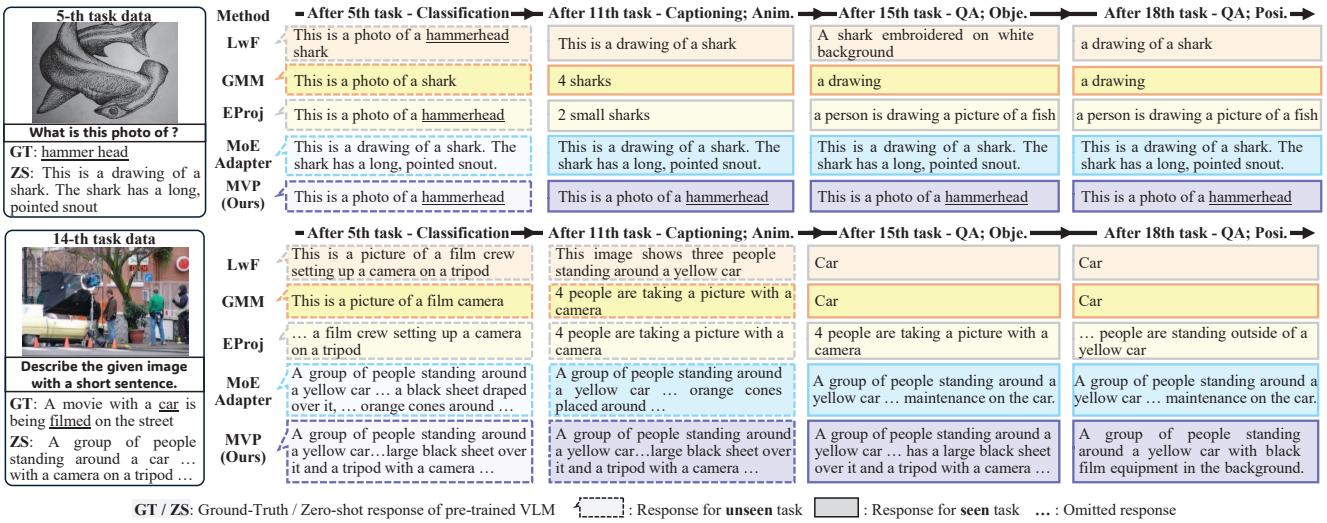


Figure 4. Qualitative results of the continual learning methods illustrating changes of responses to randomly selected samples after learning of newly emerging tasks. The proposed method generates consistent responses after learning other tasks, preserving both detailed category knowledge (top) and overall scene perception (bottom).

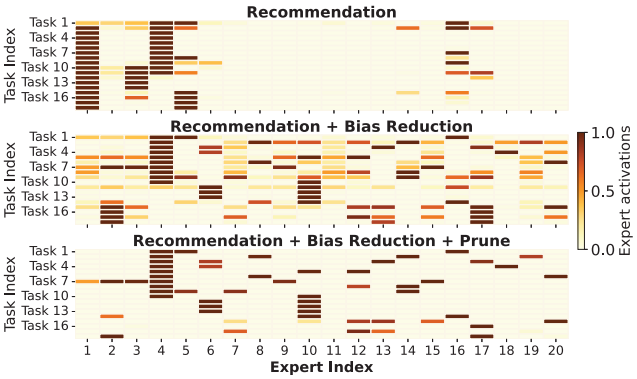


Figure 5. Results of expert activation frequency for each task by ablating the components of MVP.

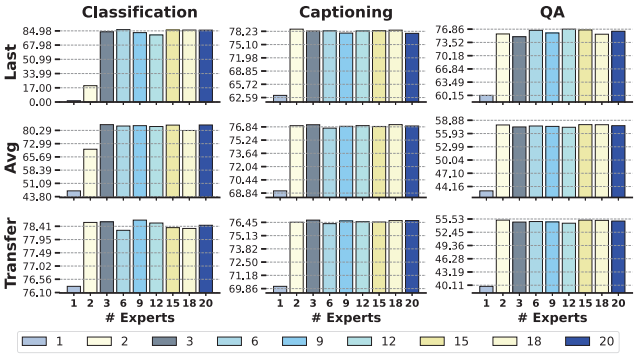


Figure 6. Results for MVP with varying numbers of experts.

activation. Notably, expert activation frequencies change at task transitions, particularly from classification to captioning (task index 11) and from captioning to question answering (task index 15).

Impact of the number of experts. We conducted additional experiments by varying the number of visual projectors in the proposed method. The results presented in Figure 6 indicate that using a small number of experts (one or two) leads to a considerable degradation in performance. In contrast, incorporating three or more experts yields performance improvements and suggests that the additional experts provide sufficient capacity for tasks with novel language instructions. Please refer to the supplementary material for additional results on varying text instructions, computational cost, and generative VLM benchmark [28].

5. Conclusion

In this paper, we have proposed a novel continual learning framework for generative vision-language models that adaptively translate visual information grounded on language instructions. We have introduced a mixture of visual projectors to leverage an appropriate visual translator based on the context of the given instruction and visual information. To mitigate the use of projectors across different contexts, we have proposed a semantic relevance-guided expert recommendation strategy that promotes the sparse activation of experts relevant to previous tasks while suppressing negative knowledge transfer. Extensive experiments on a challenging sequence of tasks, including image classification, captioning, and question answering, demonstrate that the proposed method outperforms other continual learning methods on both seen and unseen tasks.

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